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Sample Question Set - Unit 6

Course Code: CSE252
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Course Outcome: CO6
Paper: Sample Multiple Choice Questions
Maximum Marks: 40 x 1 = 40

Unit: 6
Unit Title: AI for Intelligent Systems and Robotics
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the computer vision task that locates each object present and draws a box around it.

A) Object detection
B) Image classification
C) Tokenisation
D) Clustering

Q2. Identify the element of a reinforcement learning problem that describes the situation the agent currently finds itself in.

A) The action
B) The state
C) The reward
D) The policy

Q3. State the natural language processing step that reduces words to a common dictionary base form.

A) Tokenisation
B) Vectorisation
C) Lemmatisation
D) Detection

Q4. Name the term for the purpose that lies behind a user's message to a chatbot.

A) The entity
B) The token
C) The utterance
D) The intent

Q5. Identify the reinforcement learning quantity that reduces the weight given to rewards received later.

A) The learning rate
B) The discount factor
C) The exploration rate
D) The reward signal

Q6. State what sensor fusion means for a robot.

A) Replacing every sensor with a single camera
B) Sending the sensor readings to a remote server
C) Reducing the number of sensors the robot carries
D) Combining readings from several sensors into one estimate

Q7. Explain why a vision model is trained on images that have been rotated, cropped and re-lit.

A) It teaches the model to tolerate the variation it will meet
B) It reduces the number of parameters the model needs
C) It removes the need for any labelled images
D) It makes the model run faster once deployed

Q8. Distinguish an intent from an entity in the design of a chatbot.

A) The entity is what is wanted, the intent is a detail
B) Both of the two describe the purpose of the message
C) The intent is what is wanted, the entity is a detail of it
D) Both of the two describe details within the message

Q9. Describe what a policy is in reinforcement learning.

A) A record of every reward the agent received
B) A list of the states the environment can hold
C) A rule mapping each state to the action to take
D) A measure of how well the agent has performed

Q10. Interpret an object detector returning a box with a confidence of 0.35 when the threshold is set at 0.5.

A) The detection is discarded as too uncertain
B) The detection is accepted with a warning
C) The detection is accepted and the threshold changed
D) The detector must be retrained immediately

Q11. Predict what happens to a chatbot when a user phrases a familiar request in wording it has never seen.

A) The request is answered more accurately
B) The chatbot automatically learns the new phrasing
C) The chatbot converts the request into an entity
D) The intent may not be recognised at all

Q12. Recognize the vision task described: the whole photograph is given a single label such as cat or dog.

A) Object detection
B) Image classification
C) Sensor fusion
D) Tokenisation

Q13. Explain why a learning robot must balance exploration against exploitation.

A) Without exploiting it never receives any reward
B) Exploration reduces the memory the robot needs
C) Exploitation removes the need for any sensors
D) Without exploring it never finds a better course of action

Q14. Summarize what removing stop words contributes to processing a piece of text.

A) It corrects the spelling of every word present
B) It converts each word into its dictionary form
C) It drops very common words that carry little meaning
D) It identifies the people named in the text

Q15. Distinguish a rule-based chatbot from one built on a learned language model.

A) One generalises from examples, the other matches patterns
B) One matches written patterns, the other generalises from examples
C) Both of the two match written patterns exactly
D) Both of the two generalise from training examples

Q16. Explain why the vision model of a mobile robot is often run on board rather than on a remote server.

A) The decision must be taken faster than a round trip allows
B) The model is far too large to fit on a server
C) A server cannot process image data at all
D) On-board models need no training data

Q17. Predict the effect on a robot of a reward that counts the dirt collected rather than the cleanliness of the floor.

A) It cleans the floor faster than it otherwise would
B) It may deposit dirt in order to collect it again
C) It refuses to move until the floor is already clean
D) It ignores the reward and cleans correctly anyway

Q18. Describe what a fallback response is for in a conversational system.

A) It handles a message whose intent was not recognised
B) It answers the most common question asked
C) It stores the message for later training
D) It converts the message into a set of tokens

Q19. Explain why named entity recognition is useful inside a chatbot.

A) It decides which language the message uses
B) It measures how polite the message is
C) It counts how many words the message holds
D) It extracts the specific details the request needs

Q20. Distinguish an episode from a step in reinforcement learning.

A) An episode is one action, a step a whole attempt
B) Both of the two describe a single action
C) A step is one action, an episode is a whole attempt
D) Both of the two describe a complete attempt

Q21. Apply the intersection over union measure to a predicted box of area 50 and a true box of area 70 whose overlap has area 30.

A) About 0.33

B) About 0.43
C) About 0.60
D) About 0.25

Q22. Solve the following: an agent receives rewards of 4, 2 and 8 over three successive steps with a discount factor of 0.5. Determine the discounted return.

A) 14
B) 9
C) 7
D) 5

Q23. Use a bag-of-words representation over the sentences 'robot lifts box' and 'robot drops box', and determine the size of the vocabulary.

A) 6
B) 4
C) 3
D) 2

Q24. Apply a confidence threshold of 0.6 to detections scored 0.9, 0.55, 0.7 and 0.2, and determine how many are kept.

A) 1
B) 3
C) 4
D) 2

Q25. Examine a chatbot that answers correctly for 'book a lab slot' but fails on 'I need a laboratory booking', and determine the most appropriate remedy.

A) Raise the confidence threshold of the detector
B) Reduce the number of intents the system holds
C) Add more example phrasings or match by meaning
D) Remove the entities from the training examples

Q26. Compare image classification with object detection on what each one returns for a photograph.

A) Detection returns one label, classification returns boxes
B) Both of the two return a single label per image
C) Both of the two return a box for every object
D) Classification returns one label, detection returns boxes and labels

Q27. Solve the following: a robot controller has a budget of 80 milliseconds. Its perception takes 35 and its planner 25. Determine the time left for actuation.

A) 20 milliseconds
B) 60 milliseconds
C) 10 milliseconds
D) 45 milliseconds

Q28. Apply the vocabulary of reinforcement learning to a vacuum robot that reads its position, drives to a spot and receives a score for dirt collected.

A) Position is the action, driving the state, score the policy
B) Position is the state, driving the action, score the reward
C) Position is the reward, driving the state, score the action
D) Position is the policy, driving the reward, score the state

Q29. Apply tokenisation to the sentence 'the robot moves fast' and determine the number of word tokens.

A) 3
B) 5
C) 1
D) 4

Q30. Examine a robot detector trained only on daylight photographs that fails after dark, and determine the cause.

A) The detector has too few parameters to work
B) The deployment conditions were absent from the training data
C) The confidence threshold was set too low
D) The images were stored in the wrong format

Q31. Compute the number of frames a detector can process each second when one inference takes 25 milliseconds.

A) 25
B) 4
C) 40
D) 400

Q32. Apply transfer learning to a robot vision task for which only 300 labelled images exist, and determine the usual approach.

A) Reuse a pre-trained network and retrain its final layers
B) Train a new network from the beginning on the 300
C) Abandon vision and use a rule-based method
D) Collect no more data and accept the poor result

Q33. Use an exploration rate of 0.1 and determine the proportion of actions the agent takes at random.

A) 10 per cent
B) 90 per cent

C) 1 per cent
D) 50 per cent

Q34. Compare running a vision model on board a robot with running it on a remote server, on latency.

A) The server is faster, since no model must be loaded
B) The two have identical latency in every case
C) Latency does not depend on where the model runs
D) On board is faster, since no network round trip is needed

Q35. Examine a reinforcement learning agent that keeps repeating the first adequate action it discovered, and determine the cause.

A) Its discount factor is set above one
B) Its exploration rate is too low
C) Its state representation is too detailed
D) Its reward signal is delivered too quickly

Q36. Apply the design of a reward to a delivery robot and determine the better of two definitions.

A) A score for every metre the robot travels
B) A score for every second the robot is switched on
C) A score for parcels delivered on time, with a penalty for collisions
D) A score for the number of routes it considers

Q37. Solve the following: an agent receives a reward of 10 now and 10 at the next step with a discount factor of 0.9. Determine the discounted return.

A) 20
B) 18
C) 19
D) 10

Q38. Use an intersection over union threshold of 0.5 to determine whether a detection scoring 0.42 counts as correct.

A) It counts, since any overlap is sufficient
B) It does not count, since it falls below the threshold
C) It counts, since 0.42 is close to 0.5
D) The threshold does not apply to detections

Q39. Demonstrate how a well-designed chatbot should handle a request whose intent it cannot recognise.

A) Answer with the most common intent it holds
B) Repeat the user's message back unchanged
C) Close the conversation without any reply
D) Say so, ask a clarifying question or hand over to a person

Q40. Relate the resolution of the camera images to the computation an object detector requires.

A) Higher resolution costs more computation per frame
B) Higher resolution costs less computation per frame
C) Resolution has no bearing on the computation
D) Higher resolution removes the need for a threshold

----- End of Sample Question Set -----